DAX Expressions

Function Type	DAX Output Type	Create As
Aggregation (SUM, AVG)	One result per report	Measure
Logical (IF, AND)	One value per row	Calculated Column
Text (LEFT, CONCATENATE)	One value per row	Calculated Column
Statistical (STDEV.P)	One result per report	Measure
Math (ABS, ROUND)	Depends on context	Usually a Column
Filter (CALCULATE, FILTER)	Summary values with context	Measure
Time Intelligence	Aggregate by date	Measure

Aggregation Functions

Function	Description / Purpose	Example (DAX)
SUM()	Adds up all values in a column	Total Sales = SUM(Sales_DAX[SalesAmount])
AVERAGE()	Calculates the average value in a column	Average Sales = AVERAGE(Sales_DAX[SalesAmount])
COUNT()	Counts the number of non-null values in a column	Sales Count = COUNT(Sales_DAX[SalesAmount])
COUNTROWS()	Counts the number of rows in a table	Row Count = COUNTROWS(Sales_DAX)
MIN()	Returns the smallest value in a column	Min Sales = MIN(Sales_DAX[SalesAmount])
MAX()	Returns the largest value in a column	Max Sales = MAX(Sales_DAX[SalesAmount])
DISTINCTCOUNT()	Counts the number of distinct values in a column	Unique Customers = DISTINCTCOUNT(Sales_DAX[CustomerID])

Logical Functions

Function	Purpose / Description	Example (DAX)
IF()	Checks a condition and returns one value if true, another if false	Sales Category = IF(Sales_DAX[SalesAmount] > 1000, "High", "Low")
AND()	Checks if two or more conditions are true	High Sales and Discount = AND(Sales_DAX[SalesAmount] > 1000, Sales_DAX[Discount] > 0.2)
OR()	Checks if at least one condition is true	High Sales or Discount = OR(Sales_DAX[SalesAmount] > 1000, Sales_DAX[Discount] > 0.2)
NOT()	Reverses the result of a logical test	Not High Sales = NOT(Sales_DAX[SalesAmount] > 1000)
SWITCH()	Evaluates an expression and returns a result based on multiple conditions	Sales Category = SWITCH(TRUE(), Sales_DAX[SalesAmount] > 1000, "High", Sales_DAX[SalesAmount] > 500, "Medium", "Low")

Text Functions

Function	Purpose / Description	Example (DAX)
CONCATENATE()	Joins two text strings into one	Full Name = CONCATENATE(Sales_DAX[FirstName], Sales_DAX[LastName])
LEFT()	Extracts a specified number of characters from the left side of a text string	First 3 Characters = LEFT(Sales_DAX[Product], 3)
RIGHT()	Extracts a specified number of characters from the right side of a text string	Last 3 Characters = RIGHT(Sales_DAX[Product], 3)
LEN()	Returns the length of a text string	Name Length = LEN(Sales_DAX[FirstName])
UPPER()	Converts a text string to uppercase	Uppercase Name = UPPER(Sales_DAX[FirstName])
LOWER()	Converts a text string to lowercase	Lowercase Name = LOWER(Sales_DAX[LastName])

Statistical Functions

Function	Purpose / Description	Example (DAX)
STDEV.P()	Calculates the standard deviation of a population	Sales StdDev = STDEV.P(Sales_DAX[SalesAmount])
VAR.P()	Calculates the variance of a population	Sales Variance = VAR.P(Sales_DAX[SalesAmount])

Mathematical Functions

Function	Purpose / Description	Example (DAX)
ABS()	Returns the absolute value of a number	Absolute Profit = ABS(Sales_DAX[Profit])
ROUND()	Rounds a number to a specified number of digits	Rounded Sales = ROUND(Sales_DAX[SalesAmount], 0)
POWER()	Returns the result of a number raised to a specified power	Sales Squared = POWER(Sales_DAX[SalesAmount], 2)

Filter Functions

Function	Purpose / Description	Example (DAX)
CALCULATE()	Evaluates an expression in a context modified by filters.	Sales in 2025 = CALCULATE(SUM(Sales_DAX[SalesAmount]), YEAR(Sales_DAX[Date]) = 2025)
ALL()	Removes filters from a column or table.	Total Sales All = CALCULATE(SUM(Sales_DAX[SalesAmount]), ALL(Sales_DAX[Date]))
FILTER()	Returns a table representing a subset of another table based on a condition.	Filtered Sales = CALCULATE(SUM(Sales_DAX[SalesAmount]), FILTER(Sales_DAX, Sales_DAX[SalesAmount] > 1000))

Time Intelligence Functions

Function	Purpose / Description	Example (DAX)
TOTALYTD()	Calculates the year-to-date total for an expression.	Total Sales YTD = TOTALYTD(SUM(Sales_DAX[SalesAmount]), Sales_DAX[Date])
PREVIOUSMONTH()	Returns a table with values from the previous month.	Sales Last Month = CALCULATE(SUM(Sales_DAX[SalesAmount]), PREVIOUSMONTH(Sales_DAX[Date]))
SAMEPERIODLASTYEAR()	Returns a table with values from the same period in the previous year.	Sales Same Period Last Year = CALCULATE(SUM(Sales_DAX[SalesAmount]), SAMEPERIODLASTYEAR(Sales_DAX[Date]))

Date and Time Functions

Function	Purpose / Description	Example (DAX)	Column Type
TODAY()	Returns the current date.	Current Date = TODAY()	Calculated Column / Measure
YEAR()	Extracts the year from a date.	Year = YEAR(Sales_DAX[Date])	Calculated Column
MONTH()	Extracts the month from a date.	Month = MONTH(Sales_DAX[Date])	Calculated Column
DATE()	Creates a date from individual year, month, and day values.	Custom Date = DATE(2025, 5, 4)	Calculated Column
DATEDIFF()	Calculates the difference between two dates in a specified unit (DAY, MONTH, YEAR, etc.).	Days Between = DATEDIFF(Sales_DAX[StartDate], Sales_DAX[EndDate], DAY)	Calculated Column / Measure

Date Dimension table

```
DateTable =
ADDCOLUMNS(
        CALENDAR(DATE(2015,1,1), DATE(2030,12,31)),
        "Year", YEAR([Date]),
        "Month Number", MONTH([Date]),
        "Month Name", FORMAT([Date], "MMMM"),
        "Quarter", "Q" & FORMAT([Date], "Q"),
        "Weekday", FORMAT([Date], "dddd"),
        "Year-Month", FORMAT([Date], "YYYY-MM")
)
```